

MITES AND TICKS

☉ Mites and Ticks and their Control

Arachnids are terrestrial and air-breathing arthropods include a number of parasites and pests. Three orders of this class are of economic importance viz., Acarina, Araneida and Scorpionida. Ticks and Mites belong to the order Acarina and infest a number of plants and animals. These are small sized and oval having compact bodies. Mouth parts are well adapted for biting, sucking and piercing. Chelicerae and pedipalps are small chelate and associated with the mouth-parts. The segmentation in the posterior part of the body is greatly reduced and sometimes is altogether absent. The numbers of legs are variable in larval, nymphal and adult stages. The legs are 3 pairs in the larva and nymph while 4 pairs in the adult stage. The respiration is performed by the trachea but in its absence cutaneous respiration takes place.

Mites are major agricultural pests and of considerable biological importance. They are smaller than the ticks and are predaceous, parasitic and scavengers. Some of them are microscopic in size. They are found inhabiting almost every habitat suitable for animal life viz. in fresh water, in sea-water, in hot water, in soil, on vegetation, in plant galls, under stones, in debris, on dead animals etc. Some species are parasites of insects, farm animals, birds and even on man.

Prevention and control:

- (1) The infected individuals must be isolated from the rest of the house-hold.
- (2) The body of the infected animals should be treated with a dilute mixture (5 parts water) of benzyl benzoate (68 parts), DDT (6 parts), benzocaine-ethyl p-aminobenzoate (12 parts) and Sorbiton monooleate polyxyalkalene derivative (14 parts). The mixture must not be used on the head region.
- (3) The treatment with 1% lindane or 10% sulphur ointment is also effective.

☉ *Dermanyssus gallinae* (Chicken Mite)

Phylum	Arthropoda
Class	Arachnida
Order	Acarina
Family	Dermanyssidae
Genus	Dermanyssus

The chicken mites attack preferably chickens. The other poultry is not likely to be affected badly if chickens are available. In the infested houses it causes irritation to persons or other animals. They are cosmopolitan in distribution and cause

more trouble in dry and warm countries. They live in walls, floors, cracks, ceilings of the houses during day time and at night they crawl upon the fowls. They feed normally on the blood of fowls.

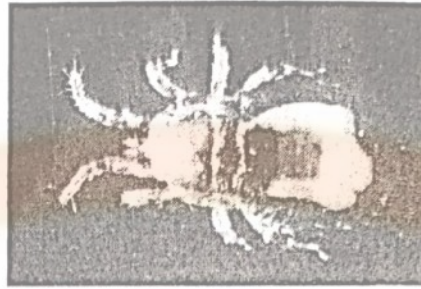


Fig 4.1: *Dermanyssus gallinae* (Chicken Mite)

General appearance: The adult chicken mites are grey coloured and 0.1 mm to 0.15 mm in length. When they are replete with blood they look bright red to black in colour. They have sharp piercing type of mouth parts. It is found that adults feed on alternate days.

Damage: The chicken mites feed on the blood of fowls by sharp piercing mouth parts at night. They never stay on poultry, so it is very difficult to know the infection of mites in primary stages. The mites are found in abundance where the roost timbers rest on their support. Affected fowl becomes droopy, listless and becomes unable for egg laying and ultimately dies.

Life history: The number of chicken mites is greatly reduced during cold weather. With the advent of spring, their activity gets increased and female lays pearly-white elliptical eggs in cracks or dry manure under the roosts. Within a period of a few weeks 25 to 35 eggs are laid and hatching occurs after 2 to 4 days producing young, six-legged nymphs. The nymphs become 8 legged when they search for a fowl and feed on blood. After third moulting the nymph becomes adult. The time taken for development from egg to an adult is about 10 days.

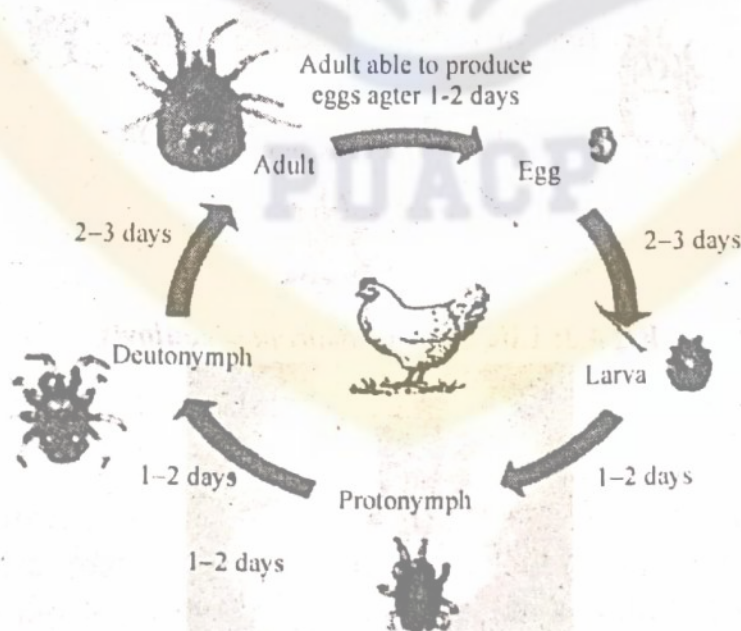


Fig 4.2: Life cycle of *Dermanyssus gallinae*

start new generation which is completed within one week. Thus a single female produces 200 to 300 adult mites.

Damage: This mite bites the individuals and hive-like eruptions appear on the body which causes itching for about a week. This also causes headache, vomiting and fever. Only adults are responsible for this infection.

Prevention and control:

- (1) People should keep themselves away from infested straw stubbles or seeds.
- (2) Before working among infested materials free use of sulphur dust in clothing should be done and work clothes should be changed soon.
- (3) Fumigation and heating at 130 °F for 6 hours protects things from infestation.
- (4) The application of cooling ointments provides slight relief from irritation.

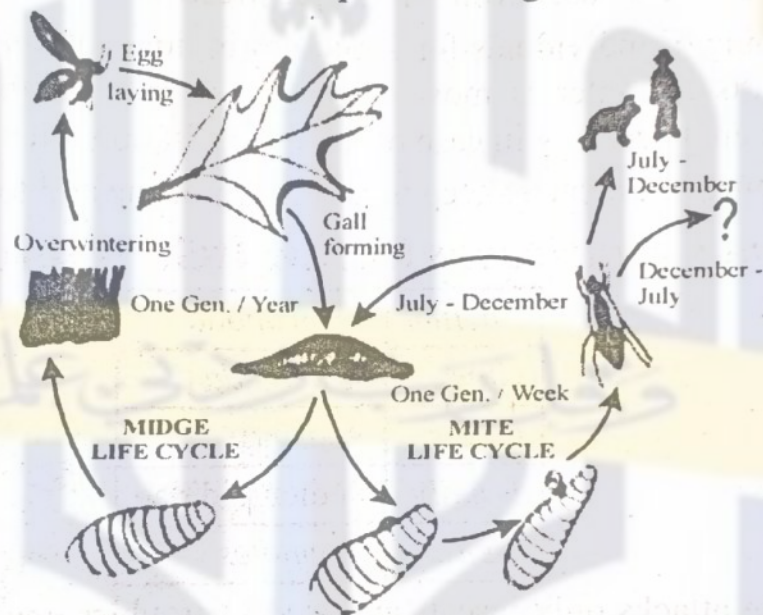


Fig 4.6: Life cycle of *Pyemotes ventricosus*

☉ ***Eutrombicula alfreddugesi* (Chiggers Mite)**

Phylum	Arthropoda
Class	Araachnida
Order	Acarina
Family	Trombidiidac
Genus	<i>Eutrombicula</i>

Chiggers attack on men, poultry, domestic animals, snakes, frogs, squirrels, rabbits etc. It is distributed in peculiar way because some areas are badly attacked, while other areas apparently similar and far distant are free. The infective stage is 6 legged larvas and not the adult.



Fig 4.7: *Eutrombicula alfreddugesi*

General appearance: The adults do not attack men and are scavengers on the excrement of other arthropods or on decaying wood. Small 6 legged mite is about 0.025 mm in diameter and cannot be seen by naked eyes. They are oval, orange coloured and run very fast.

Life history: After fertilization female lays eggs which hatch into 6 legged larval mites. The eggs are laid in the overlapping scales of snakes and turtles. The 6 legged larval stage of this mite attacks the man and causes irritation due to feeding by inserting its sharp mouth-parts in the skin. After being full fed, this 6 legged larval mite drops off from man in September and settles into the loose soil during the winter. Further, it attains the nymphal stage which alters 2-3 weeks moults and attains adult stage and remains in its earthen cells in the soil until spring. Only one generation has been recorded per year.

Damage: The 6-legged larval mite causes the appearance of scattered red blotches of varied size followed by itching for about a week. The human-being become nervous resulting into his illness. This mite injects a poisonous irritating substances.

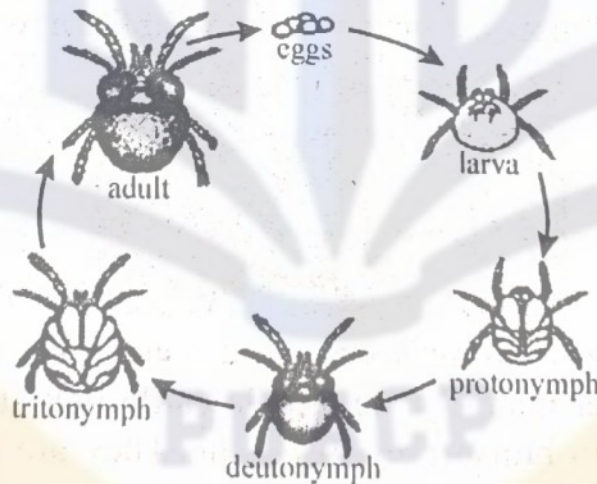


Fig 4.8: Life cycle of *Eutrombicula alfreddugesi*

Prevention and control:

- (1) Clothes treated with dimethyl phthalate, butyl phthalate or benzyl benzoate.
- (2) Dusting of 5 per cent benzyl on arms and legs and opening of clothing is quite effective to prevent the attack.
- (3) The application of cooling ointments viz., ammonia or salicylic acid in alcohol with a little olive oil prevents the infection.
- (4) The spraying or dusting with lindane at 0.25-0.5 pound, dieldrin at 0.5-1 pound, or chlordane or toxaphene, at 2-4 pound per acre is effective.

☉ Ticks

Ticks are large in size in comparison to the mites. They also belong to the order Acarina and attack animals like cattle, sheep, goat, horse etc. Unlike insects whose body is divided into 3 parts (head, thorax and abdomen) the head and thorax are fused. Some common ticks of economic importance are given here.

☉ *Boophilus microplus* (Cattle Tick)

Kingdom	Arthropoda
Class	Arachnida
Order	Acarina
Family	Ixodidae
Genus	Boophilus

The cattle tick mainly attacks the cattle but is also found on horses, mules, deer, goats and sheep. All the stages viz., larva, nymph and adult suck the blood from the host. They spread cattle fever, tick fever, texas fever from sick individuals to healthy ones. The symptoms of tick fever are high fever, reddish discoloration of urine, enlargement of spleen, loss of flesh, arched back and dropping ears. The protozoan, *Boophilus annulatus* is responsible for this disease which destroys the red corpuscles of the blood. Thus, this disease is caused by a protozoan parasite but only the ticks are responsible for the spread of this disease. The germs of texas fever are carried through the eggs of diseased ticks which infect the new individual.



Fig 4.9: *Boophilus microplus* (Canestrini)

General appearance: After maturity the female cattle tick attains a length of about 0.1 mm. It is olive-green to bluish grey in colour. They are bean-shaped with hard and wrinkled skin. The larvae are smaller provided with 3 pairs of legs, whereas, nymphs and adults have 4 pairs of legs.

Life history: The female after mating lays eggs in the soil, cracks anti crevices of buildings. A single female lays about 2000 eggs in clusters. The eggs hatch into six legged larva after 2-6 weeks. The larva attaches itself on a host. These larvae start feeding, get changed into young nymphs which attain adult stage feeding on the same infested cattle. Now mating takes place on the host's body and adults drop to the soil surface to lay eggs for another generation. Four stages viz. eggs, seed tick, nymph and adult are found in this parasite. The adults exist in two phases i.e. parasitic and non parasitic phase. Part of the life time of an adult female, egg stage and most of the seed tick (larva) stage comes under nonparasitic phase whereas a portion of the seed

tick stage, complete ' nymphal stage and a part of adult female life is parasitic phase. Single cycle needs about 60 days and 4 generations have been recorded in one year.

Damage: The soft part of host cattle is attacked by cattle tick. Mostly they attack the skin of ear and under and base of the tail. The ticks insert their head, into the skin and suck the blood due to which a number of diseases are transmitted.

Prevention and control:

1. Ticks should be removed by hand picking and killed immediately.
2. The animals affected by ticks should be sprayed by lindane dust or carbaryl/malathion 1 % per cent.
3. Rotating pasture lands should be planned due to which all seed ticks will die.
4. Dairy cattle should be dipped in 0.175 to 0.19 per cent arsenic trioxide.
5. The quarantine activities should be maintained by the Government.

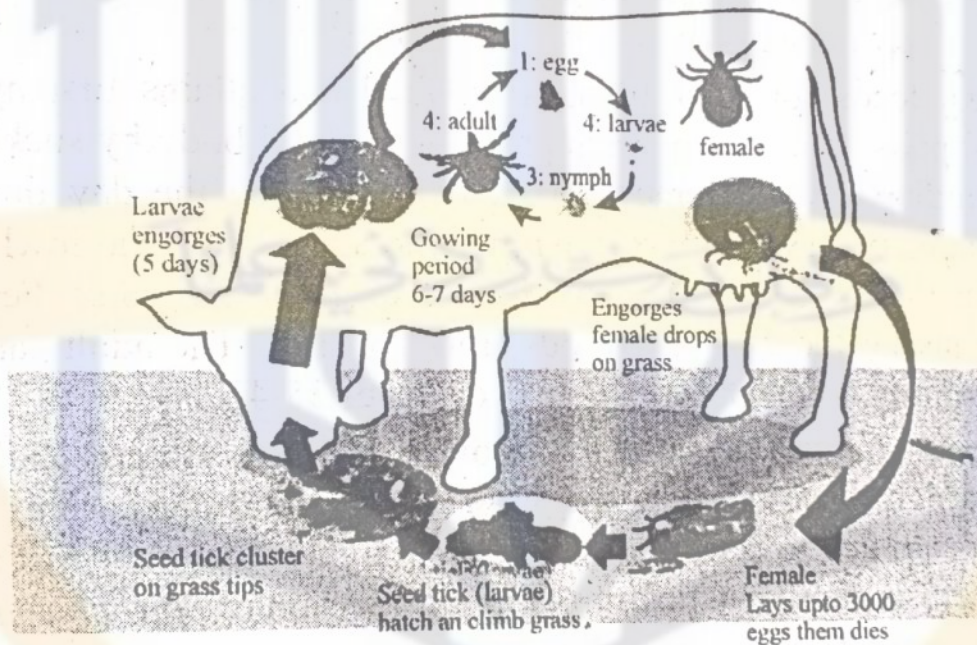


Fig 4.10: Life cycle *Boophilus microplus*

Argas persicus oken (Fowl Tick)

Phylum	Arthropoda
Class	Arachnida
Order	Acarina
Family	Argasidae
Genus	Argas

This tick is cosmopolitan in distribution and transmits spirochetes. Apart from fowls, the ducks and pigeons are also attacked by this tick. The feeding period for the nymphs and adults is only the night but larvae feed continuously till they are fully developed. The domestic mammals and men are rarely attacked.

Fig 4.11: Fowl Tick *Argas persicus* oken

General appearance: The adults are flattened, leathery and egg shaped. They are red to blue-black in colour and about 0.5 cm in length.

Life history: This tick breeds throughout the year but best breeding period is the hot dry weather. The female lays about 500-900 eggs in batches of 3-100 in cracks and crevices of fowl houses. The eggs are hatched into 6 legged larvae after 10-90 days of laying to 100 in cracks and crevices of fowl houses.

The larva feeds for 3 to 10 days on fowl and attains first nymphal stage provided with 4 pairs of legs. Now nymph feeds on fowl body by sucking the blood only at night, hiding itself in cracks and crevices during the day time. Further it moults and changes into second instar nymph who again sucks the fowl blood, drops to the soil surface and after 15 days moults and changes into adult. Female can lay eggs only after taking the meal of bloods from the host. The adult can survive for about 3 years, without food. The complete cycle takes 35-45 days.

Damage: The poultry mites attack the skin of neck, breast, thighs, under wings and suck the blood which causes weakness of legs, pale comb, droopy wings, cessation of egg-laying and death due to severe loss of blood. Highly fatal poultry diseases called as fowl spirochaetosis causes by tick through transmission of *Spirochaeta ariserina*.

Prevention and control:

1. The poultry houses should be sprayed by 3% malathion, 2% carbaryl on walls, floor and ceilings.
2. Roosts should be suspended by wires from the ceilings so that they do not touch the walls.
3. Poultry should be established only at those places where treatment of insecticides can be performed.
4. The entrance of a new fowl into the flock should be with the precaution that it is tick-free. It may be treated with 0.5% chlordane before entry.